

The Shore of Closed-String Gravity: an Exact Unitarity Boundary for the Cheung–Hillman–Remmen Graviton Family

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Abstract

The level-truncation bootstrap of Cheung, Hillman and Remmen produces a one-parameter family of graviton amplitudes containing the Virasoro–Shapiro amplitude at $\lambda = 1$, with residues that are perfect squares; partial-wave positivity was known to bound λ from below for $D \geq 9$, numerically. We derive the boundary in closed form. The near-leading even trajectory obeys $a_{n,2n-4} \geq 0 \iff D \leq T_n(\lambda)$ with $T_n(\lambda) = \frac{3(2n-3)}{n(n-2)}(\lambda^2 + (2n-2)\lambda + 1) + 2n$, verified exactly for $n \leq 14$ and against exact-rational scans. The true boundary is the envelope $\min_n T_n$: it is born at $(D, \lambda) = (9, 0)$ — explaining the onset observed by CHR — passes *exactly* through the string at $D = 23$ (so pure Virasoro–Shapiro first loses positivity on this trajectory at $D = 24$, via the level-4 spin-4 wave, matching exact scans), and straightens onto the exact asymptote $D = (12 + 4\sqrt{3})\lambda$ with active level $n^* \simeq \sqrt{3}\lambda$. Doomed-cell predictions of the envelope are confirmed by direct evaluation. We conjecture the envelope is the complete boundary and report the falsification battery it has survived.

1 Setup

Family (CHR, PRD 111, 086034): $\mu(n) = \frac{n+\lambda-1}{\lambda}$, $R(n, t) = [((1+\lambda)/2 + \lambda t)_{n-1}/((1+3\lambda)/2)_{n-1}]^2$, massless external states ($s+t+u=0$), $\lambda \geq 0$, VS at $\lambda = 1$. Partial waves: Gegenbauer coefficients at $s = \mu(n)$, $t = -\mu(n)(1-x)/2$; odd waves vanish by the $t \leftrightarrow u$ symmetry of the squared residue. All computations exact-rational; evaluator validated against CHR’s reported behavior (D=4 positivity for all λ ; onset of the lower bound at $D \geq 9$).

2 The trajectory law

Theorem 1 (Near-leading even trajectory). *For the residues above, $\lambda > 0$ and $D > 3$,*

$$a_{n,2n-4} \geq 0 \iff D \leq T_n(\lambda), \quad T_n(\lambda) = \frac{3(2n-3)}{n(n-2)}(\lambda^2 + (2n-2)\lambda + 1) + 2n. \quad (1)$$

Derivation. Write $\lambda(t - \xi(k)) = \frac{1}{2}[(n+\lambda-1)x + (2k-n)]$ at level n ; the residue is the square of $Q(x) = \prod_{k=1}^{n-1} \lambda(t - \xi(k))$ up to a positive factor. The roots $x_k = (n-2k)/(n+\lambda-1)$ are symmetric under $x \rightarrow -x$, so Q has definite parity: odd partial waves vanish identically, and the subleading coefficient q_{n-2} of Q vanishes. Hence the x^{2n-4} coefficient of Q^2 is the *pure cross*

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term $2q_{n-1}q_{n-3} < 0$ — a naive term-by-term estimate would miss the edge entirely. Only x^{2n-2} and x^{2n-4} feed the Gegenbauer mode $\ell = 2n - 4$ (parity and degree), with the exact projection ratio $\rho(\ell, D) = \frac{(\ell+1)(\ell+2)}{2(D+2\ell-1)}$; assembling the two contributions and removing positive factors yields T_n . Using $e_2(\text{roots}) = -\frac{n(n-1)(n-2)}{6(\lambda+n-1)^2}$ one obtains the equivalent closed form $T_n = \frac{3(2n-3)(\lambda+n-1)^2}{n(n-2)} - (4n-9)$.

Verification. The bracket was extracted independently by symbolic computation for every $n \leq 14$ and matches T_n identically (12/12); an independent adversarial review re-derived the theorem from scratch, and its falsification suite (published with the code) found no counterexample, including razor zeros at fractional dimensions $D^* = 271/4$ ($n=5$, $\lambda=7/2$) and $D^* = 57$ ($n=6$, $\lambda=3$) computed directly from the projection with symbolic D . Doomed-cell executions confirm predicted kill levels exactly (e.g. $(\lambda, D) = (3/2, 32)$ and $(2, 40)$: sign flip at $n = 4$ as predicted).

3 The envelope

The boundary implied by Theorem 1 is the envelope $\lambda_{\min}(D)$ defined by $D = \min_{n \geq 3} T_n(\lambda)$. Its structure:

Onset at $D = 9$. $T_n(0) = 2n + \frac{3(2n-3)}{n(n-2)}$ is minimized at $n = 3$ with $T_3(0) = 9$: the trajectory constrains nothing for $D \leq 9$ — reproducing, as a formula, the onset CHR observed numerically ($D \geq 9$).

The magic point. $T_n(1) = \frac{2(n^2+4n-9)}{n-2}$, whose integer minimum is 23, attained only at $n = 4$ (continuous minimum at $n = 2 + \sqrt{3}$). Hence $\lambda_{\min}(23) = 1$ *exactly*: the shore passes through the Virasoro–Shapiro point, and pure closed-string gravity first loses positivity on this trajectory at $D = 24$, through its level-4 spin-4 wave — matching the exact scan (first negative at $(n, \ell) = (4, 4)$ for $D = 24$, clean at $D = 22$ to depth 40).

The straight far edge. Minimizing T_n over continuous n at large λ gives the active level $n^* = \sqrt{3}\lambda$ and the exact asymptote $D = (12 + 4\sqrt{3})\lambda$; numerically $D/\lambda_{\min}$ rises $18.75 \rightarrow 18.92$ over $D = 60 \rightarrow 10^4$ against $12 + 4\sqrt{3} = 18.928\dots$, with the residual $O(1/\lambda)$ oscillation caused by the discreteness of n^* .

4 Virasoro–Shapiro thresholds

For the string itself the theorem gives the threshold sequence $D_n(1) = 2(n^2 + 4n - 9)/(n - 2) = 24, 23, 24, \frac{51}{2}, \frac{136}{5}, \dots$ — an exact refinement, on this trajectory, of the finite-level $D_{\text{crit}}(n)$ phenomenology known for Veneziano-type amplitudes. Because $D_n(1)$ grows linearly, the trajectory endangers VS only within a finite window of levels at any fixed D ; the deep asymptotic bounds of the literature arise from other (low-spin) constraints and are complementary to the exact finite-level statement proven here.

5 Conjectured completeness and its battery

Conjecture 1. *A point (λ, D) of the family is consistent with partial-wave positivity at all levels iff $D \leq \min_{n \geq 3} T_n(\lambda)$.*

The conjecture is falsifiable: a single exact negative coefficient at a point with $D \leq \min_n T_n$ refutes it. Its current battery: 494/494 agreements with exact full-spin scans (all even $\ell \leq 2n - 2$, depth 16, grid $\lambda \in [0.1, 10]$, $D \in [4, 40]$) with zero alarms of either type; two direct doomed-cell

executions at the predicted level; and an independent adversarial review whose suite (seven attack batteries) found no counterexample. The most plausible additional knife — the $\ell = 2n - 6$ trajectory — is under derivation and will either extend the boundary or strengthen the conjecture. In $D = 4$ the model predicts (and scans confirm, 40/40 to depth 14 plus the trivial bound $T_n \geq T_n(0) \geq 9 > 4$) that the entire family survives: within these assumptions, four-dimensional gravity is *not* forced to be string-like at any finite depth we can reach.

6 Discussion

[Dimension-graded rigidity of closed-string gravity; D=4 freedom; the road to predictions for gravitational-wave phenomenology; method transfer from the open-string companion paper.]

Reproducibility and AI disclosure

[Same standard as companion: exact arithmetic, published battery, adversarial review script, AI assistance disclosed in full. A sign-convention bug found and fixed during this work is documented in the public research log — including which results it did and did not affect.]

References

- [1] C. Cheung, A. Hillman and G. N. Remmen, “Uniqueness Criteria for the Virasoro-Shapiro Amplitude,” *Phys. Rev. D* 111, 086034 (2025), arXiv:2408.03362.
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